1. Introduction to Causal Inference

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Introduction* (Prof. Sanghack Lee, GSDS SNU)

The central question is why the tools of classical machine learning and statistics, which learn aspects of a *fixed* data-generating reality, are not sufficient to answer questions about actions and about worlds that were never realised.

These notes consolidate repeated (build-up) slides, expand the reasoning behind each example, and make explicit the arithmetic that the slides only state. Every slide of the source deck is covered; administrative slides are summarised briefly in Section 1. Physical PDF page numbers of the source are listed in Section 8.

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1 Course logistics and roadmap

Purpose. Record the organisation of the course and the order in which the topics of these notes appear.

Instructor. Sanghack Lee, Associate Professor, Graduate School of Data Science (GSDS), Seoul National University. Class meets Friday 9:30–12:00 (Bldg. 43-2, B102); office hours by appointment (Rm. 408, Bldg. 43). Students are encouraged to ask questions publicly on eTL and during class.

Required background.

- Mathematics: algebra, discrete mathematics (integers, graphs, statements in logic), probability and statistics, and machine learning (especially graphical models).
- The class is not purely theoretical: programming experience, a good understanding of algorithms and data structures (in particular graphs), and some experience with data analysis are expected.

Tentative schedule.

(1) Course logistics, introduction, causal models and graphs.

(2) Identification of causal effects, confounding and the back-door criterion.

(3) Adjustment criterion and do-calculus.

(4) Algorithmic identification.

(5) Potential outcomes, CATE estimation, IPW, A-IPW.

(6) Instrumental variables, matching, regression discontinuity.

(7) Difference-in-differences, synthetic control.

(8) Midterm.

(9) Partial identification and soft interventions.

(10) Causal discovery.

(11) Causal data science.

(12) Advanced topics (proximal inference, staggered DiD, synthetic DiD, DML, ...).

(13) Research trends / paper reviews; final exam.

References. There is no official textbook; reading material is distributed. Useful companions are the lectures of Ilya Shpitser, *The Effect* by Nick Huntington-Klein, course material by Kosuke Imai (Harvard) and Fan Li (Duke), and Brady Neal's video lectures. The class material is adapted from the course of Elias Bareinboim (Columbia University; Ph.D. UCLA with Judea Pearl).

Grading. Midterm 20%, final 20%, homework 60%.

2 A brief review of machine learning

Purpose. Describe what standard learning paradigms estimate, so that it becomes clear what they *cannot* estimate.

2.1 Why machine learning?

AI/ML is the design and analysis of intelligent systems. Intelligent behaviour requires knowledge, i.e. a model of the environment. Writing this knowledge down explicitly is hard and often infeasible, so it is acquired by *learning from data through algorithms.*

Learning modifies an agent's decision mechanism in order to improve performance. Since environments change, an agent must adapt to unexpected situations; learning is essential when the designer is not omniscient, and is particularly useful for messy real-world data. Three fundamental questions recur:

- How can the learning process be characterised?
- How should prior knowledge be traded off against performance?
- How robust and generalisable is the resulting system?

Applications are everywhere: generative AI (LLMs, multimodal agents, digital twins), AI for science (protein folding, drug discovery, materials), medicine (personalised treatment, surgical planning), sustainability (climate, weather, energy grids), vision (scene understanding, deepfake detection), autonomous systems (humanoid robots, drone fleets) and security (fraud detection, cyber-security agents).

2.2 The classical learning paradigm

There is an unknown reality, a true model M, that generates the world. M produces examples represented as feature vectors x, and one is interested in an ideal target function $f_M(x)$ of that reality. From training data $\{x^{(i)}\}_{i=1}^N$ one selects a hypothesis $h \in \mathcal{H}$ such that $h(x) \approx f_M(x)$ for every instance. The main challenge is generalisation to unseen instances.

Supervised learning maps a feature space $\mathcal{X}$ to a label space $\mathcal{Y}$ from labelled pairs $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$, looking for h that mimics an oracle $f : \mathcal{X} \to \mathcal{Y}$. Common hypothesis classes are non-parametric methods (decision trees, k-NN), probabilistic models (Bayesian networks), deep networks (CNNs, Transformers) and ensembles (XGBoost, LightGBM).

Unsupervised learning receives only $x \in \mathcal{X}$ and seeks a representation revealing intrinsic structure: clustering, dimensionality reduction (latent spaces), and generative modelling of the density $P(x)$. (The slides illustrate this with a phylogenetic tree, a hierarchical clustering of species.)

2.3 A probabilistic view

Both paradigms estimate features of one fixed joint distribution:

- supervised learning is essentially inference of the conditional distribution $P(y \mid x)$ (e.g. the output activation of a network or a leaf of a decision tree);
- unsupervised learning infers properties of the marginal distribution $P(x)$.

In both cases *we learn to imitate aspects of an underlying static reality.* Nothing in $P(x, y)$ tells us what would happen if the mechanism producing x were changed.

2.4 Reinforcement learning: intervention instead of observation

In reinforcement learning no static labelled sample is given. The agent *intervenes* by interacting with the environment and learns from its responses (A/B testing, robotics). The key difficulty is the exploration/exploitation trade-off. RL thus touches a new kind of quantity, the effect of an action; the course asks what is still missing from this picture, in particular how to reason about actions using data that were *not* generated by our own experiments, and how to reason about events that did not happen.

3 Conceptual foundation: the ladder of causation

Purpose. Introduce the three cognitive abilities, observing, intervening, imagining, that structure the whole course.

Human cognition can be organised in three layers:

Layer 1. Observation (sensors): the ability to observe the world and infer the "next" stage.

Layer 2. Agency: the ability to intervene in the world and change it.

Layer 3. Retrospection: the ability to reflect on what we did (and did not do) and infer new behaviours.

This cognitive stratification is formalised as **Pearl's Causal Hierarchy (PCH)**.

3.1 Inference about worlds that are not realised

The classical pipeline is

$$\text{Real world / nature} \xrightarrow{\text{data}} \text{AI/ML, statistics (inference)} \longrightarrow \text{new knowledge.}$$

For an *alternative reality* that was not realised ("what if we had changed the policy?") there are no data at all. The only route is

$$\text{knowledge} \longrightarrow \text{causal model} \xrightarrow{\text{causal inference}} \text{new knowledge about the alternative reality.}$$

A causal model carries assumptions about mechanisms that let us transport information from the observed world to unobserved ones.

3.2 Pearl's Causal Hierarchy

Level (symbol)	Activity	Question	Examples / typical models
1. Association	Seeing	What is?	What does a symptom tell us about a disease? How would seeing X change my belief in Y?
$P(y \mid x)$			(Un)supervised ML: deep nets, Bayes nets, hierarchical models, decision trees.
2. Intervention	Doing	What if?	What if I take aspirin, will my headache be cured? What if I do X?
$P(y \mid \mathrm{do}(x), c)$			Reinforcement learning, causal Bayesian networks, MDPs, POMDPs.
3. Counterfactual	Imagining,	Why?	Was it the aspirin that stopped my headache? What if I had acted differently?
$P(y_x \mid x', y')$	retrospection		Structural causal models.

Causal inference = cross-layer inference. Typical tasks take data from one layer and answer a query from a higher (or different) layer:

- *Data from Layer 1, query in Layer 2.* "Will students' performance improve if we change the class time to be earlier?" using only observational records.
- *Data from Layer 2, query in Layer 2.* "Use the findings of previous experiments to predict the outcome of a new one" (transportability, generalisation of experiments).
- *Data from Layers 1 and 2, query in Layer 3.* "Would I have won had I picked door #2 instead of door #1?"

Later lectures prove that, in general, lower layers *underdetermine* higher ones: without further assumptions, Layer 1 data cannot answer Layer 2 questions, and Layer 2 data cannot answer Layer 3 questions.

3.3 Two motivations

- **Scientific perspective:** understanding nature and human beings, using empirical data, logic and mathematics, and the scientific method.
- **Pragmatic perspective** (economics, medicine, engineering): building truly intelligent systems, and designing and implementing policies aimed at social, economic or political goals. Policies are interventions, so they live in Layer 2.

4 Correlation versus causation

Purpose. Show with real-world data that strong association is compatible with no causal relation at all.

The slides display several famous spurious correlations:

- The number of active Facebook users and the yield of 10-year Greek government bonds (2005–2011) rise together (BusinessWeek, "Correlation or causation?", Dec. 2011).
- Fresh lemons imported to the USA from Mexico and total US highway fatality rate, 1996–2000, have $R^2 = 0.97$ (negative slope).
- Organic food sales and diagnosed autism prevalence, 1998–2007, have $r = 0.9971$ ($p < 0.0001$).
- The divorce rate in Maine correlates with the per-capita consumption of margarine (2000–2009).

Each pair is driven by common time trends or coincidence rather than by a mechanism linking them. The xkcd cartoon (no. 552) jokes: "I used to think correlation implied causation. Then I took a statistics class. Now I don't." "Sounds like the class helped." "Well, maybe." The real question is the converse one: *if correlation does not imply causation, then what does?* The answer developed in this course is: correlation *plus* causal assumptions, encoded in a model.

5 Example 1: Simpson's paradox in a 2×2 table

Purpose. Demonstrate that the same data support opposite conclusions depending on the causal story, so that no purely statistical rule decides which table to use.

Notation: X = took the drug, $\neg X$ = did not; Y = survived (recovered), $\neg Y$ = did not; F = female, $\neg F$ = male (Simpson 1951; Pearl 2000, Ch. 6).

All	Y	$\neg Y$	total	rate
drug (X)	20	20	40	50%
no drug ($\neg X$)	16	24	40	40%
Σ	36	44	80	

Male ($\neg F$)	Y	$\neg Y$	total	rate
drug	18	12	30	60%
no drug	7	3	10	70%
Σ	25	15	40	

Female (F)	Y	$\neg Y$	total	rate
drug	2	8	10	20%
no drug	9	21	30	30%
Σ	11	29	40	

Within each sex the drug is worse,

$$P(Y \mid F, X) = 0.2 < 0.3 = P(Y \mid F, \neg X),$$
$$P(Y \mid \neg F, X) = 0.6 < 0.7 = P(Y \mid \neg F, \neg X),$$

yet in the aggregate it is better: $P(Y \mid X) = 0.5 > 0.4 = P(Y \mid \neg X)$.

Why this is arithmetically possible. By the law of total probability,

$$P(Y \mid X) = P(Y \mid X, \neg F)P(\neg F \mid X) + P(Y \mid X, F)P(F \mid X).$$

Among drug takers $P(F \mid X) = 10/40 = 0.25$, while among non-takers $P(F \mid \neg X) = 30/40 = 0.75$. Hence

$$P(Y \mid X) = 0.75 \cdot 0.6 + 0.25 \cdot 0.2 = 0.50,$$
$$P(Y \mid \neg X) = 0.25 \cdot 0.7 + 0.75 \cdot 0.3 = 0.40.$$

The aggregate rates average the stratum rates with *different weights*; men, who recover more often regardless of treatment, are over-represented among drug takers. The reversal is not a paradox of probability, only of intuition.

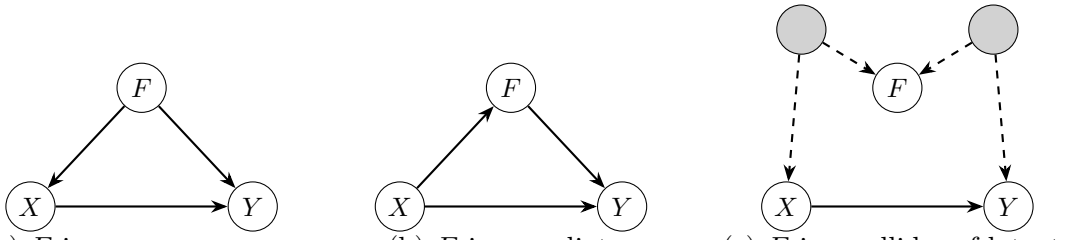
Which table should the physician consult? If sex affects both who takes the drug and recovery, the stratified table should be used, and the population-level effect is obtained by weighting the strata by the *population* distribution of sex, $P(F) = P(\neg F) = 40/80 = 1/2$:

$$\textstyle\sum_f P(Y \mid X, f)P(f) = \frac{1}{2}(0.6) + \frac{1}{2}(0.2) = 0.40,$$
$$\textstyle\sum_f P(Y \mid \neg X, f)P(f) = \frac{1}{2}(0.7) + \frac{1}{2}(0.3) = 0.50,$$

so the drug lowers recovery by 10 percentage points. (This formula is the back-door adjustment of Lecture 4.)

5.1 Would the result be different with a different story?

Keep the numbers but change the causal story behind them. The slides draw three graphs:



(a) F is a common cause (b) F is a mediator (c) F is a collider of latents

- **(a)** $F \to X$, $F \to Y$, $X \to Y$. F (sex) confounds the treatment–outcome relation. Consult the stratified tables; the drug is harmful.
- **(b)** $X \to F \to Y$, $X \to Y$. Suppose the third variable were not sex but something the drug changes (e.g. blood pressure). Stratifying on it would block part of the drug's effect; the aggregate table gives the total effect and the drug is beneficial.
- **(c)** $X \leftarrow U_1 \to F \leftarrow U_2 \to Y$, $X \to Y$ with U_1, U_2 unobserved. F is a collider; conditioning on it *creates* a spurious association between X and Y. The aggregate table is correct.

The data are identical in the three scenarios, yet the right answer differs. The decision therefore depends on knowledge outside the data, the causal structure, which is exactly what causal models represent.

6 Example 2: a "trivial" data analysis

Purpose. Show that a perfect fit does not reveal the direction or meaning of a dependence.

Firefighters (X)	Size of fire (Y)
10	100
40	400
60	600
200	2000

The points lie exactly on $Y = 10X$, so the regression $Y = \beta X + \beta_0$ fits perfectly with $\beta = 10$, $\beta_0 = 0$. X and Y are perfectly correlated, but does X cause Y? Read causally, $\beta = 10$ would say that each additional firefighter adds ten units of fire, which is absurd: the size of the fire determines how many firefighters are dispatched ($Y \to X$). *After all, what is the meaning of β?* A regression coefficient is a property of the joint distribution; whether it measures the effect of changing X depends on the causal mechanism, which the fit is silent about.

7 What is going on here?

Both examples ask for something that is not in the joint distribution: the effect of an action. Answering such questions requires (i) a formal language for causal assumptions, (ii) a semantics connecting assumptions with data and queries, and (iii) inference rules deciding when and how a causal query is computable from available data. Lectures 2–7 build exactly this machinery: structural causal models, causal graphs, identification, the back-door and adjustment criteria, do-calculus, and algorithmic identification.

8 Source coverage

The source deck has 58 physical pages (34 numbered slides, many with incremental build-ups). Build-up stages are consolidated into a single exposition.

Topic	Physical pages
Logistics, schedule, references, grading	1–13
Review of machine learning (paradigm, supervised, unsupervised, RL)	14–31
Three layers and Pearl's Causal Hierarchy	32–42
Scientific/pragmatic perspective, correlation vs. causation	43–49
Simpson's paradox	50–54
Firefighter example and closing question	55–58